MILITARY SPECIFICATION

INTERMEDIATE FREQUENCY AMPLIFIER AM-6608( )/ULR-17(V)

This specification is approved for use by the Electronics Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 This specification covers the requirements for the construction and testing of the Intermediate Frequency Amplifier AM-6608( )/ULR-17(V) (hereafter referred to as the Precision IF unit) which is part of a matched channel receiving subsystem. The Precision IF unit AM-6608 shall suitably amplify and filter the 455 kHz IF signal from the R-1849( )/ULR-17(V) Radio Receiver or the CV-3168( )/ULR-17(V) Frequency Converter. The Precision IF unit described by this specification is to be used in conjunction with a R-1849( )/ULR-17(V) Radio Receiver, a Control Receiver Unit C—9376, a J-3176( )/ULR-17(V) Interconnecting Box, a CV-3168( )/ULR-17(V) Frequency Converter, and a J-3177( )/ULR-17(V) Interconnecting Box to complete a matched channel subsystem. (See Figure 1).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on the date of invitation for bids or requests for proposal, form a part of the specification to the extent specified herein.

Beneficial comments, (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, ERADCOM, Fort Monmouth N.J. 07703, ATTN: DELET-DP, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 5820

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SPECIFICATIONS

FEDERAL

NN-P-71	Pallets, Material Handling, Wood, Stringer Construction, 2 Way and 4 Way (Partial)
QQ-S-781	Strapping, Steel, Seals
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Shipping, Fiberboard
PPP-B-640	Boxes, Fiberboard, Corrugated, Triple-Wall
PPP-F-320	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes
PPP-S-760	Strapping, Nonmetallic (and Connectors)
PPP-T-97	Tape, Pressure-Sensitive Adhesive, Filament Reinforced

MILITARY

MIL-P-116	Preservation-Packaging, Method of
MIL-P-11268	Parts, Materials, and Processes Used In Electronic Equipment
MIL-M-13231	Marking of Electronic Equipment
MIL-F-14072	Finish For Ground Electronic Equipment
MIL-P-19644	Plastic Molding Material (Polystyrene Foam, Expanded Bead)
MIL-P-26514	Polyurethane Foam, Rigid or Plastic, For Packaging

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-147	Palletized and Containerized Unit Loads 40 Inch x 48 Inch Pallet, Skids, Runner or Pallet Type Base

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STANDARDS, MILITARY (Cont'd)

MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461A	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics and Measurement of
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-810	Environmental Tests Methods
MIL-STD-1188	Commercial Packaging of Supplies and Equipment
MIL-STD-1472	Human Engineering Design Criteria for Military Systems

DRAWINGS

ELECTRONICS COMMAND

DLSM-B-783850	AM-6608( )/ULR-17(V) Data List
SM-D-2014510	AM-6608( )/ULR-17(V) Precision IF Unit
SM-A-689133	Acceptance Test Procedure for Precision IF (A5)

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.

3.2 Construction. The equipment shall be constructed in accordance with the requirements of this specification and the drawings listed in the USAECOM Data List DLSM-B-783850. The Precision IF AM-6608( )/ULR-17(V) top assembly drawing number is SM-D-2014510.

3.3 Parts, materials, and process; general. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein shall apply.

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3.3.1 Finish. The AM-6608 shall be finished in accordance with MIL-F-14072 and the drawings listed on DLSM-B-2014510.

3.4 Marking.

3.4.1 General. Marking shall conform to MIL-M-13231 and MIL-STD-130 (See 4.4). Front panel marking shall be Group I as described in MIL-M-13231.

3.4.2 Visibility. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.4.3 Serial numbers. Each AM-6608 shall have a serial number.

3.5 Performance.

3.5.1 General characteristics. The Precision IF, AM-6608, is an IF amplifier with carefully controlled phase and gain characteristics. Two or more AM-6608 amplifiers, when used with the associated supplemental RF units, CV-3168, and HF receiver, R-1849, provide a multichannel phase and gain matched receiving capability.

3.5.2 Selectivity. The bandwidth of the AM-6608 shall be as follows:

	Low	High
3dB bandwidth (max)	453.45 kHz	456.75 kHz
12dB bandwidth (max)	452.5 kHz	458.5 kHz
60dB bandwidth (max)	448.25 kHz	461.75 kHz

3.5.3 Ripple. The RF ripple shall be consistent with the amplitude response requirements of 3.5.4.

3.5.4 Amplitude response. The amplitude response, relative to the value at 455 kHz, shall be within the limits specified in Table I.

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TABLE I. AM-6608 Frequency Response
(Relative to value at 455 kHz)

FREQ (kHz)	MIN. GAIN (dB)	NOMINAL GAIN (dB)	MAX GAIN (dB)
454.0	-1.90	-1.60	-1.30
454.1	-1.55	-1.30	-1.05
454.2	-1.25	-1.00	-0.75
454.3	-1.05	-0.80	-0.55
454.4	-0.85	-0.60	-0.35
454.5	-0.65	-0.40	-0.15
454.6	-0.50	-0.25	0.00
454.7	-0.40	-0.15	+0.10
454.8	-0.30	-0.05	+0.20
454.9	-0.30	-0.05	+0.20
455.0	0	0	0
455.1	-0.30	-0.05	+0.20
455.2	-0.30	-0.05	+0.20
455.3	-0.35	-0.10	+0.15
455.4	-0.45	-0.20	+0.05
455.5	-0.55	-0.30	-0.05
455.6	-0.75	-0.50	-0.25
455.7	-0.85	-0.60	-0.35
455.8	-1.05	-0.80	-0.55
455.9	-1.25	-1.00	-0.75
456.0	-1.50	-1.20	-0.90

3.5.5 Gain.

3.5.5.1 Absolute gain. The overall gain of the AM-6608, at 455 kHz, with the attenuators set for 90 dB loss, shall be 9 $\pm$ 0.5 dB.

3.5.5.2 Gain control. Gain shall be controlled by external electrical signals. The gain shall be variable in 1 dB steps. Ninety nine dB of gain control shall be available.

3.5.5.3 Attenuator tolerance. The allowable range of attenuator loss for each attenuator section shall be as specified below. The attenuator loss is defined as the loss when the attenuator set for maximum attenuator minus the loss when the attenuator section is set for maximum attenuation.

Nominal attenuator value	Min Loss (dB)	Max Loss (dB)
10 dB	9.90	10.10
8 dB	7.92	8.08
4 dB	3.95	4.05
2 dB	1.95	2.05
1 dB	0.95	1.05

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3.5.6 Phase. The phase versus frequency characteristics of the AM-6608 shall correspond to the requirements of Table II over the temperature range of -4°F to 160°F and also, to the requirements of Table III over the temperature range -40°F to -4°F.

TABLE II. Phase-Frequency Characteristics
(-4°F to +160°F)

FREQ	PHASE		
	MIN	NOMINAL	MAX
454.0	111.1	113.1	115.1
454.1	99.8	101.8	103.8
454.2	88.5	90.5	92.5
454.3	77.4	79.4	81.4
454.4	66.2	68.2	70.2
454.5	55.0	57.0	59.0
454.6	43.8	45.8	47.8
454.7	33.5	34.5	35.5
454.8	22.1	23.1	24.1
454.9	10.6	11.6	12.6
455.0	0	0	0
455.1	-12.6	-11.6	-10.6
455.2	-24.1	-23.1	-22.1
455.3	-35.6	-34.6	-33.6
455.4	-48.0	-46.0	-44.0
455.5	-59.4	-57.4	-55.4
455.6	-70.8	-68.8	-66.8
455.7	-82.0	-80.0	-78.0
455.8	-93.2	-91.2	-89.2
455.9	-104.4	-102.4	-100.4
456.0	-115.5	-113.5	-111.5

3.5.7 Differential delay. The differential delay over the frequency range 454.000 kHz to 456.000 kHz shall not exceed 35 microseconds.

3.5.8 Inputs.

3.5.8.1 IF input. The IF input of the AM-6608 shall have a center frequency of 455.000 kHz. The input impedance shall be not less than 33 ohms and not greater than 75 ohms.

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3.5.8.2 Gain control input. The gain control inputs shall be low power TTL compatible.

3.5.9 Outputs.

3.5.9.1 IF output. The saturated IF output power shall be +5dBm (min) at 455.000 kHz into 50 ± 5 ohm load.

3.5.9.2 AGC IF output. The AGC IF output shall be 225 ± 35 millivolts into a 1000 ± 100 ohm load when the IF output (see 3.5.9.1) is 0dBm into a 50 ± 5 ohm load. Saturated output shall be + 5dBm (min).

TABLE III. Phase Frequency Characteristics
($-40^{\circ}F$ to $-4^{\circ}F$)

FREQ	PHASE		
	MIN	NOMINAL	MAX
454.0	110.1	113.1	116.1
454.1	98.8	101.8	104.8
454.2	87.5	90.5	93.5
454.3	76.4	79.4	82.4
454.4	65.2	68.2	71.2
454.5	54.0	57.0	60.0
454.6	42.8	45.8	48.8
454.7	32.5	34.5	36.5
454.8	21.1	23.1	25.1
454.9	9.6	11.6	13.6
455.0	0	0	0
455.1	-13.6	-11.6	-9.6
455.2	-25.1	-23.1	-21.1
455.3	-36.6	-34.6	-32.6
455.4	-49.0	-46.0	-43.0
455.5	-60.4	-57.4	-54.4
455.6	-71.8	-68.8	-65.8
455.7	-83.0	-80.0	-77.0
455.8	-94.2	-91.2	-88.2
455.9	-105.4	-102.4	-99.4
456.0	-116.5	-113.5	-110.5

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3.5.10 Input power. The AM-6608 shall meet the requirements specified in 3.5.9, when supplied with input power meeting the following requirements:

Input Voltage	Ripple
+ 5 ± 0.1 volts	5 millivolts peak to peak (max)
+ 2.5 ± 0.025 volts	5 millivolts peak to peak (max)
- 12 ± 0.24 volts	5 millivolts peak to peak (max)

3.6 Environmental. The AM-6608 herein after called equipment shall meet the following service conditions without degradation in specified performance of 3.5 unless specified herein.

3.6.1 Temperature altitude. The equipment shall be operable without degradation in specified performance at altitudes up to 30,000 feet above sea level and shall withstand air transportation at altitudes up to 50,000 feet above sea level. The equipment shall be capable of continuous operation without degradation in specified performance at ambient temperatures as low as -40 degrees F and as high as +131 degrees F with intermittent operation to +160 degrees F. The equipment shall withstand exposure to ambient air temperature as low as -70 degrees F and as high as +160 degrees F. Testing shall be in conformance with the test requirements of 4.9.1.

3.6.2 Humidity. The equipment shall be operable without degradation in specified performance, and shall sustain no physical damage, during and after prolonged exposure to relative humidities as high as 100 percent at all ambient temperatures to +85 degrees F and relative humidities corresponding to a dew point of +85 degrees F at all temperatures between +85 degrees F and +120 degrees F, and low relative humidity of 5 percent at +120 degrees F.

3.6.3 Sand and dust. The equipment shall withstand, in both operating and nonoperating conditions, exposure to fine sand and dust particles with wind speeds of up to 20 knots. After testing as specified in 4.9.3, the equipment shall meet the specified performance requirements.

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3.6.4 Salt fog. In both operating and nonoperating conditions the equipment shall withstand the salt fog test of 4.9.4. Upon completion of test, the AM-6608, when examined visually with the aid of a 10-power magnifier, shall show no evidence of degradation, such as flaking, pitting, blistering or loosening of finish or metal surfaces; or exfoliation of metal. There shall be no clogging or binding of controls or other moving parts after subjecting the AM-6608 to the 48-hour salt fog exposure period or after subjection to the subsequent 48-hour drying period.

3.6.5 Fungus. After the fungus test of 4.9.5, the equipment shall be visually examined, using a 10-power magnifier. The AM-6608 shall show no more than six minute unrelated spots, each no greater than 0.015 square inch in area, of sparse microbial growth (see 6.3.3) as evidenced by growth colonization (see 6.3.2) which includes branching (see 6.3.1) and sporulation (see 6.3.4) on or within each cubic foot, or fraction thereof, of equipment assembly volume. Isolated instances of partial tubular germination (see 6.3.5) shall not be included in this evaluation.

3.6.6 Vibration and shock. The equipment shall withstand vibration and shock induced during field transport, by military vehicle, by rough handling, and mounted in military aircraft. Vibration isolation will be provided by the systems integrator when installed in the airframe.

3.6.7 Bench handling. The equipment shall withstand bench handling (see paragraph 4.9.7).

3.6.8 Explosive atmosphere. The equipment shall not cause ignition of an ambient-explosive gaseous mixture with air when operating in such an atmosphere.

3.7 Electromagnetic interference (EMI). The equipment shall comply with the following emission and susceptibility requirements of Notice 4, MIL-STD-461A:

CE01	CS01	RE02	RS03.2
CE03	CS02	RE02.1	
CE04	CS06		
CE05			

3.8 Interchangeability. Like units, assemblies, subassemblies, and replaceable parts shall conform to requirement 7 of MIL-STD-454 (see 4.8).

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3.9 Burn-In. Each unit on order shall meet the requirements herein, without subsequent processing, after subjection to burn-in as specified in 4.6.

3.10 Workmanship. The AM-6608 shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs.

a. In MIL-P-11268

General requirements for plastic materials and plastic parts.
Wiring and cabling, Including:
- Slack
- Protection
- Clearance
- Splicing
- Connections, general
- Grounding, general
- Shielding on wire and cable

Flux and cleaning agent for soldering.
Process for soldering.
Cleaning
Riveting
General requirements for securing of parts.
Cleaning after welding.
Mechanical operation of controls.
Process for welding.
Self-tapping screws.

b. In this specification:

3.3.1 Finish

3.11 Systems safety engineering.

3.11.1 Personnel hazards. Personnel hazards shall be kept to a minimum through compliance of configuration changes and parts selection with Requirement 1 of MIL-STD-454. Compliance with these requirements will be verified through a visual inspection (see 4.10).

3.11.2 Exposed edge rounding. Exposed edges shall be rounded to a minimum radius of 0.04 inch (1 mm), and exposed corners to a minimum of 0.05 inch (1.3 mm).

3.11.3 Radioactive materials. Radioactive materials shall not be used (e.g., luminous dials/markings, electron tubes, surge arrestors and lenses).

3.12 Human factors engineering. Design configuration as affecting operation, maintainability and handling shall be maintained through compliance of configuration criteria to the requirements of MIL-STD-1472.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection requirements specified herein are classified as follows:

a. First Article inspection (does not include packaging). (see 4.5).

b. Inspection covered by subsidiary documents. (see 4.4).

c. Quality conformance inspections.

(1) Quality conformance inspection of equipment before packaging. (see 4.5).

(2) Quality conformance inspection of packaging. (see 4.11).

4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish four first article units of Precision IF AM-6608( )/ULR-17(V).

4.3.2 First article inspection. The first article inspection shall consist of inspections specified in subsidiary documents covering the items listed in 4.4, and the inspections specified for Group A, Group B, and Group C (see Table V, VI, and VII, respectively). The inspection shall be performed in the following order: 4.4, Group A and Group B for all units and designated units for specific Group C tests in the order shown in Table IV.

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TABLE IV. Environmental Test for First Article Inspection

Inspection (Note 1)	Unit 1	Unit 2	Unit 3	Unit 4
Vibration		1		
Temperature/altitude	1			
Shock, bench handling				1
Humidity	2			
Fungus (Note 2		2		
Salt Fog (Note 2)			1	
Shock				2
Sand and dust (Note 2)				3
Explosive atmosphere			2	

NOTE 1: Other First Article inspection may precede, follow, or be interspersed between the tests.

NOTE 2: The equipment shall be thoroughly washed, cleaned, dried, and refurbished, if necessary, before proceeding with subsequent tests.

4.4 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification.

ITEM	WHERE REQUIRED
Finish, Protective	3.3.1
Marking	3.4
Parts, Materials and Processes	3.3

4.5 Quality conformance inspection of equipment before packaging. Each unit of each lot of equipment shall be burned-in prior to inspection. The contractor shall perform the inspection specified in 4.4 and 4.5.1 thru 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

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4.5.1 Group A inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table V. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subject to sampling inspection utilizing the general inspection levels, and the AQL's indicated in Table V.

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in an order satisfactory to the Government.

TABLE V. Group A Inspections.

Inspection	Reg. para.	Test para	AQL Major	AQL Minor
Visual and Mechanical	3.10	4.10	1.0%	4.0% *
Electrical				
Amplitude response	3.5.4	4.7.3	1.0%	
Absolute gain	3.5.5.1	4.7.4	1.0%	
Gain control	3.5.5.2	4.7.5	1.0%	
Attenuator tolerance	3.5.5.3	4.7.6	1.0%	
Phase	3.5.6	4.7.7	1.0%	
Differential delay	3.5.7	4.7.8	1.0%	
IF output	3.5.9.1	4.7.11	1.0%	
AGC IF output	3.5.9.2	4.7.12	1.0%	
Input power	3.5.10	4.7.13	1.0%	

*All electrical defects shall be considered major.

4.5.2 Group B inspection. Group B inspection is required on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met Group A inspection. This inspection shall conform to Table VI and to the special inspection level of Table I of MIL-STD-105.

4.5.2.1 Group B sampling plans. The Group B sampling plan for the ALQ's listed in Table III shall be S-4. All defects shall be considered major.

4.5.2.2 Order of inspection within Group B. Group B inspection shall be performed in any order satisfactory to the Government.

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TABLE VI. Group B Inspections.

Inspection	Req. para.	Test para.	AQL
Interchangeability	3.8	4.8	6.5%
Selectivity	3.5.2	4.7.1	1.0%

4.5.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. Sample units shall be selected in accordance with 4.5.3.1. The inspection shall consist of the inspection specified in Table VII.

4.5.3.1 Sampling for Group C inspection. Four samples of the complete equipment shall be selected at random, 1 sample for each Sub-group I through and including Sub-group IV in Table VII. The first samples shall be selected at the start of production from the first quality conformance inspection lot. These samples shall constitute the Group C requirement for the first 25 units produced. Thereafter, 4 samples of the complete equipment shall be selected at random, 1 sample for each Sub-group I through Sub-group IV in Table VII from every 75 units and fraction thereof produced when the fraction is greater than or equal to one-half (1/2). When the fraction is less than one-half (1/2), the previous group upon which sampling was performed shall include the fractional quantity. Salt fog and fungus shall not be repeated after the first production lot Group C inspection.

TABLE VII. Group C Inspections.

Inspection	Req. para.	Test para.
Sub-Group I		
Temperature/Altitude	3.6.1	4.9.1
Vibration	3.6.6	4.9.6
Sub-Group II		
Humidity	3.6.3	4.9.2
Sub-Group III		
Shock	3.6.7	4.9.7
Sub-Group IV		
Sand and Dust 1/	3.6.3	4.9.3
Salt Fog 1/	3.6.4	4.9.4
Fungus 1/	3.6.5	4.9.5
Explosive Atmosphere	3.6.8	4.9.8
EMI	3.7	4.9.9

1/ The equipment shall be thoroughly washed, cleaned, dried, and refurbished after this inspection before proceeding with subsequent inspection.

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4.5.3.2 Order of inspection within Group C. Group C inspection shall be performed in any order acceptable to the Government.

4.5.3.3 Group C failure. Action required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of conforming Group C sample units. Unless otherwise specified sample units which have been subject to and passed Group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.6 Burn-in. Every equipment shall be subjected to a 100 hour burn-in. The last 24 hours of the burn-in shall be failure free. If an equipment fails during the last 24 hour failure free period, it shall be repaired, returned to test, and the test shall continue until a 24 hour failure free period has been achieved.

4.6.1 Compliance. Units passing burn-in shall be submitted to the electrical measurements of Group A and B inspection without benefit of repair, cleaning, realignment or other changes, unless otherwise specified herein. If after successful completion of burn-in, the equipment fails either of the electrical measurements in Group A or Group B inspection, the equipment shall be reworked and resubmitted to the last 24 hours of the burn-in period and shall successfully complete the 24 hour failure free period of burn-in prior to being resubjected to failure free Groups A and B inspections.

4.6.2 Conditions preceding burn-in. A thermal survey and vibration survey shall be conducted in accordance with MIL-STD-781. The results of the required time to stabilize at the low temperature shall determine the length of the temperature cycle to be conducted during burn-in.

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4.6.3 Test conditions. Each equipment shall be subjected to the following test conditions during burn-in:

Temperature -30°C to 55°C (-22°F to 131°F)

Temperature Cycling - Temperature cycle shall be: time to stabilize at low temperature as determined by the thermal survey, time to stabilize at high temperature, six (6) hours at the high temperature.

Vibration -2.2 G ± 10% peak acceleration value at any nonresonant frequency between 20 and 60 cps measured at the mounting points on the equipment. The duration of vibration shall be at least 10 minutes during each hour of operating time.

Equipment on-off cycling - Equipment off during the cooling cycle and on during the heating cycle.

Input Voltage - Nominal voltage of 110 volts +5% -2%.

Input Voltage Cycling - The input voltage shall be maintained at 110% nominal for one-third of the equipment "ON" cycle, at the nominal value for the second one-third of the equipment "ON" cycle and at 90% for the final one-third of the equipment "ON" cycle. This cycling procedure is to be repeated continuously throughout burn-in.

4.6.4 Failure definition. A failure is defined as the inability of the equipment to perform its required function. This includes any event which causes departure from performance as required by this specification and the contractors approved test procedure.

4.7 Performance.

4.7.1 Selectivity. The AM-6608 shall meet the requirements of 3.5.2 when tested in accordance with 5.2 of Test Procedure SM-A-689133.

4.7.2 Ripple. When the AM-6608 is tested in accordance with 5.3 of Test Procedure SM-A-689133 it shall meet the requirements of 3.5.3.

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4.7.3 Amplitude response. The amplitude response of the AM-6608 shall meet the requirements of 3.5.4 when tested in accordance with 5.3 of Test Procedure SM-A-689133.

4.7.4 Absolute gain. The AM-6608 shall meet the requirements of 3.5.5.1 when tested in accordance with 5.1 of Test Procedure SM-A-689133.

4.7.5 Gain control. The AM-6608 shall meet the requirements of 3.5.5.2 when tested in accordance with 5.4 of Test Procedure SM-A-689133.

4.7.6 Attenuator tolerance. The AM-6608 shall meet the requirements of 3.5.5.3 when tested in accordance with 5.4 Test Procedure SM-A-689133.

4.7.7 Phase. The AM-6608 shall meet the requirements of 3.5.6 when tested in accordance with 5.6 of Test Procedure SM-A-689133.

4.7.8 Differential delay. The AM-6608 shall meet the requirements of 3.5.7 when tested in accordance with 5.6 of Test Procedure SM-A-689133.

4.7.9 IF input. Meeting the requirements of 3.5.4 and 3.5.5.1 shall satisfy the requirements of 3.5.8.1.

4.7.10 Gain control input. Meeting the requirements of 3.5.5.3 shall satisfy the requirements of 3.5.8.2.

4.7.11 IF output. The AM-6608 shall meet the requirements of 3.5.9.1 when tested in accordance with 5.1.6 of Test Procedure SM-A-689133.

4.7.12 AGC IF output. The AM-6608 shall meet the requirements of 3.5.9.2 when tested in accordance with 5.5 of Test Procedure SM-A-689133.

4.7.13 Input power. With the input voltages set to the values specified in 3.5.10, the AM-6608 shall meet the requirements of 3.5.2 through 3.5.9.

4.8 Interchangeability.

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4.8.1 Inspection for dimensional interchangeability. The dimensions listed below shall be gauged or measured to determine conformance to the physical interchangeability requirement of 3.8. When a listed dimension is not within specified or design limits it shall be considered a major defect.

a. External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

d. Location of connectors, keying pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major units.

e. Size and form of special threads.

4.8.2 Electrical interchangeability. All like modules and replaceable parts shall be physically and functionally interchangeable without modification of such items or of the equipment as determined by the test specified in 3.8 or 4.8.1. However, electrical retuning and realignment of modules during production assembly operations may be expected and are permissible.

4.9 Environment. The AM-6608 shall be tested in accordance with the methods and procedures of MIL-STD-810 given below. The limits of allowable performance degradation shall be as specified in 3.5 of this specification, and shall form the criteria for passing or failing each environmental test unless stated otherwise. Successful completion of any particular test shall be considered evidence of meeting the corresponding environmental condition.

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4.9.1 Temperature - altitude. The equipment shall be subjected to the test of Method 504 of MIL-STD-810 for Class 4 equipment with the following changes and additions:

a. The temperature for Step 1 shall be -70 degrees F.

b. The temperature for Steps 2 and 3 shall be -40 degrees F.

c. The procedure shall include a one hour exposure (nonoperating) to a pressure of 3.4 inches of Hg simulating air transportation at 50,000 feet at ambient temperature.

Prior to testing the equipment shall meet full specification performance for the measurements specified in 3.5.

4.9.2 Humidity. The equipment shall be subjected to the test of Method 507, Procedure II of MIL-STD-810. In Step 6, measurements shall be taken during the first 5 hours of each cycle. Inability of the AM-6608 to meet the requirements of 3.5 shall constitute failure of this test.

4.9.3 Sand and dust. The equipment shall be tested in accordance with Method 510, Procedure I of MIL-STD-810. Prior to testing the equipment shall meet full specification performance for the measurements specified in 3.5.

4.9.4 Salt fog. The equipment shall be tested in accordance with Method 509, of MIL-STD-810. Operation of the equipment shall not be required, and failure criteria shall be limited to corrosion of finishes and metals only. Such corrosion shall be defined as any visible degradation of the test items surfaces that can be attributed to flakey, pitted, blistered, or otherwise loosened finish or metal surface.

4.9.5 Fungus. The equipment shall be subjected to the fungus test specified in Method 508, Procedure I of MIL-STD-810. There shall be abundant growth colonization (6.3.2) on 50 percent or more of the area of the control item after 14 and 28 days. No cleaning of the equipment is permitted for 72 hours prior to the fungus test. Handling, prior to and during testing, shall be accomplished without contamination of the equipment. Inability of the equipment to meet the requirements of 3.6.5 shall constitute failure of this test.

4.9.6 Vibration. The equipment shall be subjected to the vibration test of MIL-STD-810, Method 514, Procedure I, Curve C, Part 1 from Figure 5.14.2-2. Prior to testing, the equipment shall meet full specification performance for the measurements of 3.5.

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4.9.7 Bench handling. The equipment shall be tested in accordance with Method 516, Procedure V, of MIL-STD-810. Prior to testing the equipment shall meet full specification performance for the measurements specified in Table V.

4.9.8 Explosive atmosphere. The equipment shall be subjected to the test of Method 511, Procedure I of MIL-STD-810.

4.9.9 EMI. The equipment shall be tested for compliance with the requirements of 3.7. The test shall be in accordance with MIL-STD-462, Notice 3, as implemented by a contractor prepared, government approved EMI test plan.

4.10 Visual and mechanical inspection. The AM-6608 shall be examined for the defects listed in MIL-STD-252. An inspection shall also be performed to verify compliance with those portions of 3.11. (Systems Safety Engineering) which can be determined visually.

4.11 Quality conformance inspection of packaging. Packaging shall be inspected in accordance with MIL-P-116 to determine conformance to the requirements of 5 herein. The rough handling tests will be performed only when specified (See 6.2).

4.12 Human engineering verification. Maintenance of design configuration to the requirements of 3.12 shall be verified by visual and mechanical inspection.

5. PACKAGING.

5.1 Preservation. Preservation shall be level A, B, or Commercial as specified (See 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Precision IF AM-6608( )/ULR-17(V) shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Precision IF AM-6608( )/ULR-17(V) shall be dried in accordance with the applicable procedures of MIL-P-116.

5.1.1.3 Preservative application. No preservative required.

5.1.1.4 Unit packing. Unit packing shall be in accordance with the methods/submethods prescribed in MIL-P-116 as specified herein.

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5.1.1.4.1 Technical literature. Each technical literature shall be unit packed submethod 1C-1.

5.1.1.4.2 Precision IF AM-6608( )/ULR-17(V). Each Precision IF AM-6608( )/ULR-17(V) shall be individually unit packed method III as follows: Cushion on all surfaces with cells or pads or both fabricated of fiberboard conforming to PPP-F-320, type CF, class weather-resistant, variety SW, grade W5c, designed and fabricated to protect all projections and absorb the shock of impact in handling and transit. As an alternate, cushioning material may be fabricated of plastic molding material conforming to MIL-P-19644 or polyurethane foam conforming to MIL-P-26514. Place the cushioned unit within a close-fitting fiberboard box conforming to PPP-B-636, W5C. Place the technical literature, unit packed as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

5.1.2 Level B. Cleaning, drying, preservative application and unit packing shall be as specified in 5.1.1.

5.1.3 Commercial preservation. Preservation shall be as specified in MIL-STD-1188.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2). Shipping containers for level A and B shall be capable of stacking and supporting superimposed loads during shipment and storage without damaging the container(s) or its contents.

5.2.1 Level A. A quantity of Precision IF AM-6608( )/ULR-17(V), unit packed as specified in 5.1, shall be packed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type 1, finish A.

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5.2.1.1 Unitization. Palletization shall be required when: containers specified in 5.2.1 do not require skids; quantities per destination comprise a pallet load; and container size permits use of one of the pallet patterns of MIL-STD-147. A quantity of containers packed as specified in 5.2.1, except that container strapping may be omitted, shall be placed on a pallet, load type 1, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type 1, finish A, or shrink film.

5.2.2 Level B. A quantity of Precision IF AM-6608( )/ULR-17(V), unit packed as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, class 2, style E, or PPP-B-636, type CF, class weather-resistant. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by pressure-sensitive filament tape banding or non-metallic strapping conforming to PPP-T-97 and PPP-S-760, respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to NN-P-71, group I or II woods. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type 1, finish A, or shrink film. When a single unit is shipped to a single destination no packing is required; the unit container shall serve as the shipping container. Reinforcing shall be as specified above.

5.2.2.1 Unitization. Palletization shall be required when quantities per destination comprise a pallet load. A quantity of Precision IF AM-6608( )/ULR-17(V), unit packed as specified in 5.1, shall be placed on a pallet, load type 1, conforming to MIL-STD-147. The pallet shall conform to NN-P-71, type IV, group I or II woods. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping conforming to QQ-S-781, type 1, finish A, or shrink film.

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5.2.3 Commercial packing. Packing shall be in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial marking. In addition to any special marking required by the contract or order, interior packs and exterior shipping containers shall be marked in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The AM-6608 is one of high frequency components of Receiver Set, Countermeasure AN/ULR-17(V) which provides a monitoring capability of signals in the high frequency band of 0.5 to 30.0 MHz.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Marking and shipping of equipment.

c. Place of final inspection.

d. Technical literature required.

e. The C-9376, R-1849, CV-3168, J-3177 and J-3176 shall be Government furnished property (loaned).

f. Level A, Level B, or Commercial preservation and packing (see Section 5).

g. When rough handling test is required (see 4.11).

6.3 Definitions.

6.3.1 Branching. Branching is a connected arrangement of filaments (hyphae) formed by shoots or secondary stems growing from the main stem or filament (hypha)

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6.3.2 Growth colonization. Growth colonization is a mass of individual plants, generally of one species, living together; or a group of hyphae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitutes abundant growth.

6.3.3 Microbial growth. Microbial growth is the growth of very minute organisms. Such organisms when present in large numbers may provide a colony visible to the naked eye.

6.3.4 Sporulation. Sporulation is the formulation of minute unicellular reproductive or dormant bodies, called spores.

6.3.5 Tubular germination. Tubular germination is partial growth by the production of hyphae, which are tubular shaped fungal filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

6.3.6 Exfoliation. Exfoliation is corrosion along the grain boundaries of the metal resulting in the peeling or separating, or both, of successive layers of the metal. The appearance resembles loose book pages or onion skin peeling.

6.4 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

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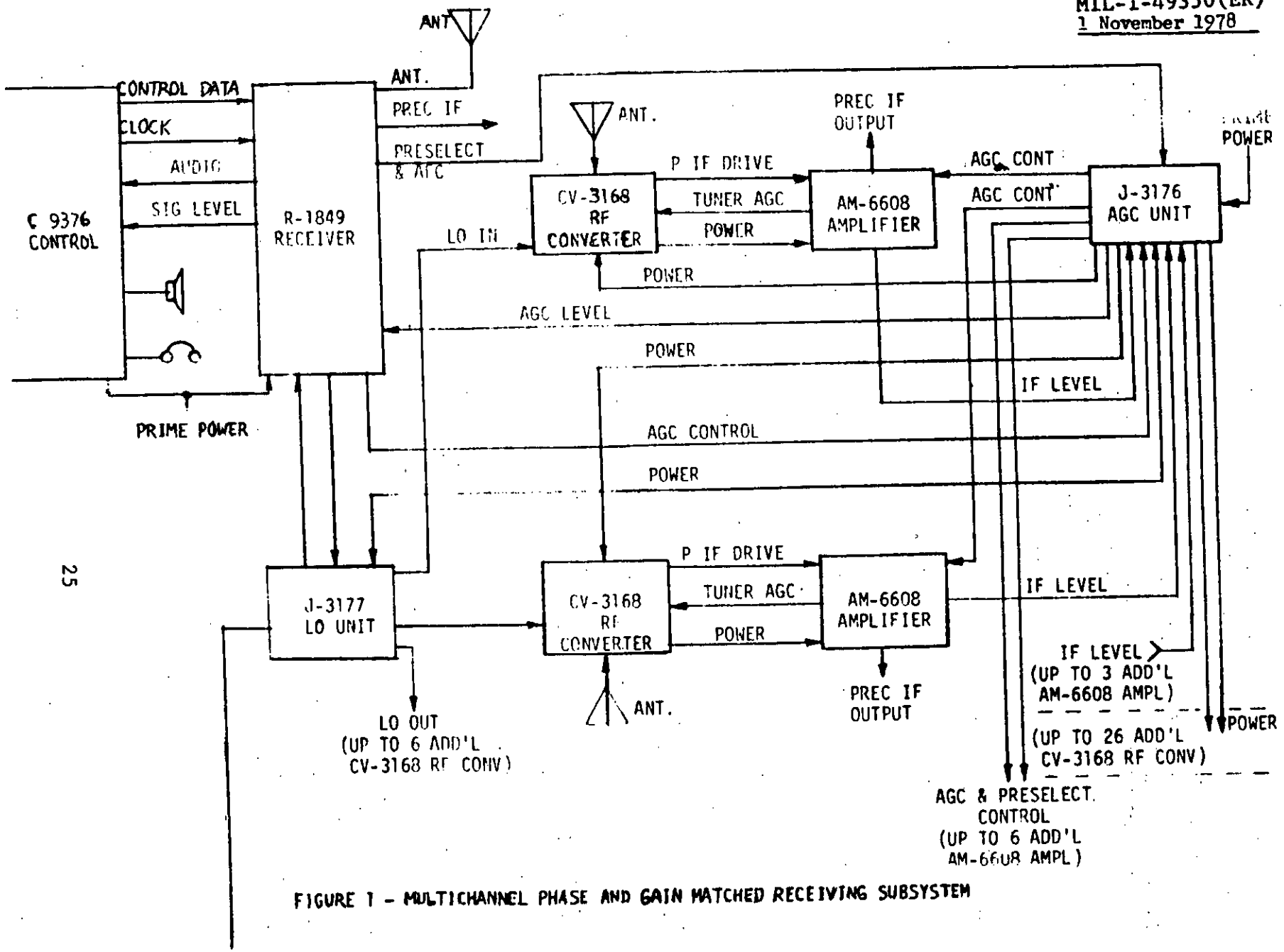


FIGURE 1 - MULTICHANNEL PHASE AND GAIN MATCHED RECEIVING SUBSYSTEM

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DOCUMENT IDENTIFIER AND TITLE MIL-I-49350(ER) INTERMEDIATE FREQUENCY AMPLIFIER AM-6608( )/ULR-17(V)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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